

Total No. of Questions : 10]

SEAT No. :

P6521

[Total No. of Pages : 2

[5870]-2002

T.E. (Computer Engineering)

**PRINCIPLES OF CONCURRENT AND DISTRIBUTED
PROGRAMMING**

(2012 Pattern) (Semester - II) (310249)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1)** a) Explain Platform Model in OpenCL in detail with suitable diagram. [6]
b) Explain the structure of YACC file. [4]

OR

- Q2)** a) Explain how to count task dependency. [6]
b) Explain how to send and receive a message in MPI. [4]

- Q3)** a) Explain the mechanism of process migration. [6]
b) Compare GPU and CPU. [4]

OR

- Q4)** a) Describe two ways of creating threads in Java. [6]
b) Explain : [4]

- | | |
|-----------------|---------------|
| i) YYTEXT | ii) YYLVAL |
| iii) YYPARSE() | iv) YYWRAP() |

- Q5)** a) Explain the following terms with respect to operating system: [8]
i) System image
ii) Autonomy
iii) Fault Tolerance Capability
b) Compare workstation model with workstation server model. [8]

OR

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- Q6)** a) Explain the processor pool model along with advantages and disadvantages of it. [8]
b) Write short note on DCE Components. [8]

- Q7)** a) Explain Xen architecture with neat diagram. [8]
b) What is the need of Virtualization? [4]
c) Explain para virtualization. [4]

OR

- Q8)** a) Explain Dom0 and DomU Communication in Xen. [8]
b) Differentiate between Virtual Machine and Distributed Machine. [4]
c) Explain full virtualization. [4]

- Q9)** a) Write a CUDA program for addition of two matrices. [8]
b) What is meant by single node and multiple nodes systems, with respect to multi CPU and multi GPU? [10]

OR

- Q10)** a) What are different types of data storage in CUDA? [8]
b) Explain the concept of mobile computing with respect to the following points: [10]
i) Mobile computing classification
ii) Advantages
iii) Future trends
iv) Limitations
v) Security issues before mobile computing

